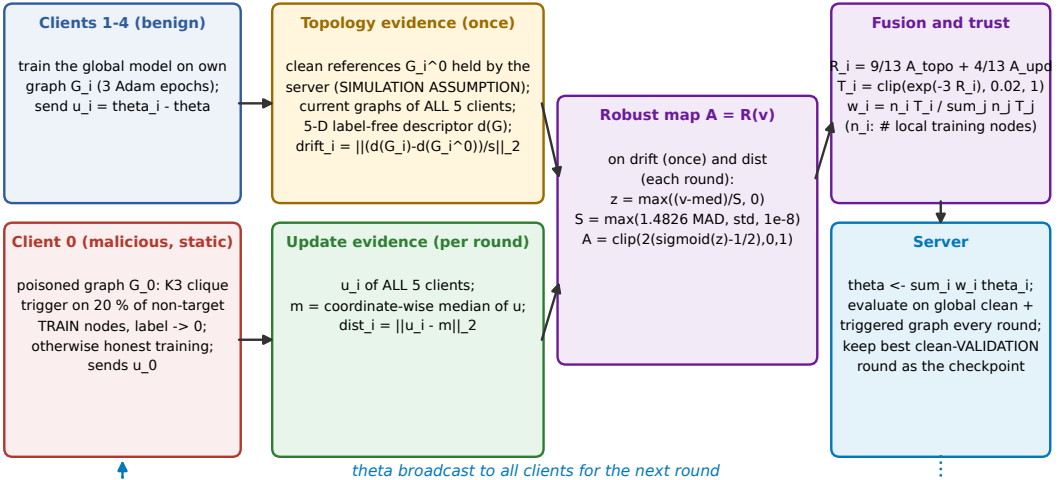
ETD-FGL as implemented (simulation: K = 5 clients, M = 1 attacker, IID partition)
Clients 1-4 (benign)
train the global model on own graph G_i (3 Adam epochs); send u_i = theta_i - theta
Topology evidence (once)
clean references G_i^0 held by the server (SIMULATION ASSUMPTION); current graphs of ALL 5 clients; 5-D label-free descriptor d(G); drift_i = ||(d(G_i)-d(G_i^0))/s||_2
Robust map A = R(v)
on drift (once) and dist (each round): z = max((v-med)/S, 0) S = max(1.4826 MAD, std, 1e-8) A = clip(2(sigmoid(z)-1/2),0,1)
Fusion and trust
R_i = 9/13 A_topo + 4/13 A_upd T_i = clip(exp(-3 R_i), 0.02, 1) w_i = n_i T_i / sum_j n_j T_j (n_i: # local training nodes)
Server
theta <- sum_i w_i theta_i; evaluate on global clean + triggered graph every round; keep best clean-VALIDATION round as the checkpoint
Client 0 (malicious, static)
poisoned graph G_0: K3 clique trigger on 20 % of non-target TRAIN nodes, label -> 0; otherwise honest training; sends u_0
Update evidence (per round)
u_i of ALL 5 clients; m = coordinate-wise median of u; dist_i = ||u_i - m||_2
theta broadcast to all clients for the next round
Logged per client and round: A_topo, A_upd, R, T, w. Reported at the selected checkpoint: test accuracy, ASR, clean target rate, attack effect.